

Lecture 3

Probability and probability distributions

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LPO.8800: Statistical Methods for Education Research



Introduction



Describing univariate distributions

- Measures of central tendency, variability, skewness, position
- Data transformations

1



Probability concepts



In descriptive statistics, all of the outcomes are known—we are simply finding useful ways of summarizing them. When using statistics for inferential purposes, one has to think about *hypothetical* outcomes. Probability provides the tools to do this.

- Probability is a *model* of the **data generating process** (i.e., the process generating the outcomes we observe)
- This model tells us the relative frequency of outcomes in an unobserved *population*.
- Like any model, its usefulness depends on the application at hand and its ability to provide a description of the real world.



2

Basic concepts:

- A **random process** is the realization of outcomes that are unknown in advance (*ex ante*) but where the possible outcomes are known.
- The **sample space** is the set of all possible outcomes of a random process (“everything that could happen”).
- An **event** is a subset of the sample space. It may be a specific outcome or a collection (set) of outcomes.
- **Probability** is the likelihood of an event occurring. Often denoted $P(\text{event})$



3

Examples of sample spaces, using set notation $\{ \}$:

- coin toss: $\{H, T\}$
- 2 coin tosses: $\{HH, HT, TH, TT\}$
- six-sided die roll: $\{1, 2, 3, 4, 5, 6\}$
- day of weather: $\{\text{rain, no rain}\}$
- amount of rain in a day $\{0 \rightarrow ?\}$
- years of education completed by an adult $\{1 - 20+\}$

The sample space may consist of *discrete* or *continuous* outcomes. It is sometimes denoted S or Ω .



4

Event

Examples of events:

- 2 coin tosses: $\{H \text{ on either toss}\}$
- die roll: $\{2\}$, $\{1, 2\}$, or $\{\text{even number}\}$
- amount of rain in a day: $\{2 \text{ inches}\}$ or $\{1 \text{ to } 3 \text{ inches}\}$
- years of education completed by an adult: $\{16 \text{ or more}\}$

We can use notation to refer to events. For example, event $A = \{1, 2\}$ or $B = \{\text{Heads on either toss}\}$



5

Rules of probability



Basic axioms of probability

The probability model has three basic axioms, or postulates. They are these simple assumptions:

1. $P(A) \geq 0$ for some event A
2. $P(S) = 1$ for the sample space S
3. If $A_1, A_2, \dots, A_N$ are **mutually exclusive** events, then:

$$P(A_1 \cup A_2 \cup \dots \cup A_N) = P(A_1) + P(A_2) + \dots + P(A_N) = \sum_{i=1}^N P(A_i)$$



With **equally likely** mutually exclusive discrete outcomes, the probability of event A is:

$$P(A) = \frac{\text{number of outcomes in } A}{\text{number of possible outcomes}}$$

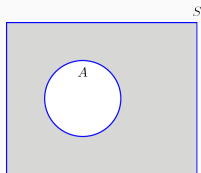
Examples: $P(H)$ on a single coin toss = $1/2$. $P(2 \text{ or } 3)$ on a die roll = $2/6$.



7

Probability rules: applying the axioms

- If $P(A)$ is the probability that event A occurs, $P(\sim A) = 1 - P(A)$ is the probability that A does *not* occur.
- $\sim A$ is the **complement** of A (something other than A occurs, the gray area below)
- $P(A) + P(\sim A) = 1$



8

Examples:

- 2 coin tosses: $S = \{HH, HT, TH, TT\}$
 - $P(\sim \text{coin comes up heads once}) = (1 - (2/4)) = 0.50$
 - $P(\sim \text{two heads}) = (1 - (1/4)) = 3/4 = 0.75$
- die roll events: $\{2\}$, $\{1, 2\}$, or $\{\text{even number}\}$
 - $P(\sim 2) = (1 - (1/6)) = 5/6 = 0.833$
 - $P(\sim \text{even number}) = 3/6 = 0.50$



9

Probability rules: applying the axioms

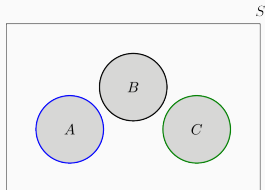
- When A and B are two *mutually exclusive* events, then
 $P(A \cup B) = P(A) + P(B)$
- $\cup$ is the symbol for the **union** of A and B—read as events A or B occurring.
- Mutually exclusive means there is no overlap in the events, so
 $P(A \cap B) = 0$
- $\cap$ is the symbol for the **intersection** of A and B—read as events A and B *both* occurring.



10

Probability rules: applying the axioms

Mutually exclusive events A, B, and C:



$$P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

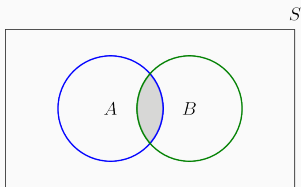
$$P(A \cap B) = P(A \cap C) = P(B \cap C) = 0$$

11



Probability rules: applying the axioms

- When A and B are *not* mutually exclusive events, then
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
- That is, since A and B overlap, we must subtract the part common to both:



12



Non-mutually exclusive events: examples

Suppose you would like to enroll in an easy class (A) or a class that is graded easily (B). What is the probability that a random class is *either* easy or graded easily?

- Assume $P(A) = 0.25$
- Assume $P(B) = 0.35$
- Assume $P(A \cap B) = 0.20$, that is 20% of classes are both easy *and* graded easily. Then:
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A \cup B) = 0.25 + 0.35 - 0.20 = 0.40$



13

Non-mutually exclusive events: examples

In the U.S. Census, race and ethnicity are separate questions. In 2010:

- 74.8% of the population is white (alone or in combination with others): $P(W) = 0.748$
- 16.3% of the population is Hispanic or Latino: $P(H) = 0.163$
- 9.5% of the population is white *and* Hispanic: $P(W \cap H) = 0.095$.
Then:
- $P(W \cup H) = P(W) + P(H) - P(W \cap H)$
- $P(W \cup H) = 0.748 + 0.163 - 0.095 = 0.816$



14

Conditional probability

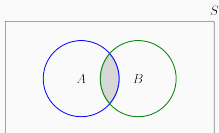
Sometimes we are interested in knowing the probability that an event will occur *given that we know another event has occurred*. Consider events A and B . There are 4 possibilities:

$$A \cap B$$

$$A \cap \sim B$$

$$\sim A \cap B$$

$$\sim A \cap \sim B$$



15



Conditional probability

Now suppose we know B occurred. This means either of the following occurred:

- $A \cap B$

- $\sim A \cap B$

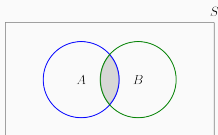
In only one of these cases does A occur.

16



Let $P(A|B)$ be the **conditional probability** that event A occurs given that event B occurred.

$$P(A|B) = \frac{P(A \cap B)}{P(A \cap B) + P(\sim A \cap B)} = \frac{P(A \cap B)}{P(B)}$$



17



Probability rules: conditional probability

Example: In an election, 20 of 50 women and 12 of 40 men oppose a policy. A ballot drawn at random indicates opposition. What is the probability it was cast by a man? $P(\text{man}|\text{oppose})$

$$\begin{aligned} P(\text{man}|\text{oppose}) &= \frac{P(\text{man} \cap \text{oppose})}{P(\text{oppose})} \\ &= \frac{12/90}{32/90} \\ &= 0.375 \end{aligned}$$

What is the probability it was cast by a woman? $P(\text{woman}|\text{oppose})$

$$\begin{aligned} P(\text{woman}|\text{oppose}) &= \frac{P(\text{woman} \cap \text{oppose})}{P(\text{oppose})} \\ &= \frac{20/90}{32/90} \\ &= 0.625 \end{aligned}$$

18



Probability rules: conditional probability

Example: in a population of 100 people, 90 are vaccinated and 10 are not. 4 are infected with COVID-19, including 3 vaccinated and 1 unvaccinated. Among those infected, what is the likelihood they were vaccinated?

$$\begin{aligned}P(\text{vaccinated}|\text{infected}) &= \frac{P(\text{vaccinated} \cap \text{infected})}{P(\text{infected})} \\&= \frac{3/100}{4/100} \\&= 0.75\end{aligned}$$

Seems alarming! But this conditional probability gets larger the greater the share of the population that is vaccinated. It's probably not the conditional probability you're looking for, if you're interested in whether vaccines "work."



19

Probability rules: conditional probability

Given a vaccination, what the probability of being infected with COVID-19? How does this compare to the unvaccinated?

$$\begin{aligned}P(\text{infected}|\text{vaccinated}) &= \frac{P(\text{vaccinated} \cap \text{infected})}{P(\text{vaccinated})} \\&= \frac{3/100}{90/100} = 0.0333 \\P(\text{infected}|\text{unvaccinated}) &= \frac{P(\text{unvaccinated} \cap \text{infected})}{P(\text{unvaccinated})} \\&= \frac{1/100}{10/100} = 0.100\end{aligned}$$

These conditional probabilities are much more informative about the benefits of vaccination.



20

Probability rules: conditional probability

Can find unconditional and conditional probabilities from a 2×2 table with counts:

Table 1: Job approval of President Biden, March 2021

	Approve	Disapprove	Total
Men	805	838	1,643
Women	802	492	1,294
Total	1,607	1,330	2,937

$$P(\text{approve}) = 1607/2937 = 54.7\%$$

$$P(\text{approve}|\text{woman}) = 802/1294 = 62.0\%$$

$$P(\text{woman}) = 1294/2937 = 44.1\%$$

$$P(\text{woman}|\text{approve}) = 802/1607 = 49.9\%$$



21

Bayes' Rule

A useful application of conditional probabilities: using knowledge that B occurred to revise or “update” the probability that A occurred:

$$\begin{aligned}P(A|B) &= \frac{P(A \cap B)}{P(B)} \\&= \frac{P(B \cap A)}{P(B)} \\&= \frac{P(B|A)P(A)}{P(B)}\end{aligned}$$

Useful for moving between conditional probabilities, and between the *prior* $P(A)$ and *posterior* $P(A|B)$. Note the third line uses the definition of $P(B|A)$.



22

A common application of Bayes' Rule is in screening for a particular disease. For example, let the event A = person has HIV, and B = person has a positive ELISA test (blood screening). Assume:

- $P(A) = 1\%$ (overall prevalence of HIV, adult males)
- Test is positive 98% of the time when HIV is present (2% false negatives)
- Test is negative 94% of the time when HIV is not present (6% false positives)

A person might ask: what is the probability I have HIV given a positive test? $P(A|B)$



23

Bayes' Rule example

Applying Bayes' Rule:

$$\begin{aligned} P(A|B) &= \frac{P(B|A)P(A)}{P(B)} \\ &= \frac{(0.98)(0.01)}{P(B)} \end{aligned}$$

What is the probability of a positive test $P(B)$?

$$P(B) = \underbrace{(0.98)(0.01)}_{P(B|A)P(A)} + \underbrace{(0.06)(0.99)}_{P(B|\sim A)P(\sim A)} = \mathbf{0.0692}$$

So:

$$P(A|B) = \frac{(0.98)(0.01)}{0.0692} = \mathbf{14.2\%}$$

Note: the underbraces come from $P(B) = P(B \cap A) + P(B \cap \sim A)$ and the definition of conditional probability.



24

In the population, 1% have HIV. Conditional on a positive blood test, 14.2% have HIV. The new evidence (B) updates the probability of A .

Notably, the test is positive 98% of the time for those with HIV: $P(B|A)$. Still, only 14.2% of those who screen positive have HIV. Why? HIV has a low *overall* incidence in the population: $P(A)$.

A physician might be very attentive to $P(B|A)$. But the patient should be more attentive to $P(A|B)$.



25

Independence

- If A and B are **independent events**, this means the probability that A occurs does not depend on whether B occurred, and vice versa.
- Put another way, knowing whether B occurred provides *no information* about the probability that event A occurred (and vice versa).
- When A and B are independent events, $P(A|B) = P(A)$
- From the definition of conditional probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$.
- Thus when A and B are independent, $P(A \cap B) = P(A) \times P(B)$. This is the *multiplicative rule*.



26

Independence: examples

Are these independent events?

- Rolling a die: event A is “rolling an even number,” and event B is “rolling a 2 or 4.” Is it true that $P(A \cap B) = P(A) \times P(B)$?
- Tossing a coin twice: event A is “getting H on the first toss,” and event B is “getting H on the second toss.”
- Drawing a random person from the U.S. Census: event A is “the person is white,” and event B is “the person is Hispanic or Latino.”
- Event A is having HIV; event B is having a positive blood test.
- Event A is having had COVID-19 vaccination; event B is having a COVID infection.



27

Independence: examples

The outcomes of sampling *with replacement* are independent events. Recall that 16.3% of the population in the 2010 Census was Hispanic.

- Draw three random people from the Census. What is the probability that all three are Hispanic?
- $P(HHH) = 0.163 \times 0.163 \times 0.163 = 0.0043$
- Draw three random people from the Census. What is the probability that the first is Hispanic but the other two are not?
- $P(H, \sim H, \sim H) = 0.163 \times (1 - 0.163) \times (1 - 0.163) = 0.114$



28

What's the difference between "independent" and "mutually exclusive"?

- If A and B are mutually exclusive events, this means that if A occurred, B did not occur.
- In this case A and B are clearly not independent. Knowing A occurred tells you all you need to know about whether B occurred.
- With independent events, the probability that A occurred is unaffected by whether B occurred.



Conceptions of probability



Where do probabilities come from?

- **Classical interpretation:** the assumption of equally likely outcomes provides a logical probabilities. Ex: $P(H)$ on a single coin toss is 0.50. $P(2 \text{ or } 3)$ on a die roll will be $2/6 = 1/3$.
- **Frequentist interpretation:** if a random trial is repeated a large (infinite) number of times under identical conditions, $P(A)$ is the fraction of times A occurs.
 - We might assume the probability that a newborn baby is female is 0.5 (or 50%). In fact, over millions of repeated observations, it is closer to 0.48 in the U.S.
- **Subjective interpretation:** probabilities based on intuition



30

Probability as relative frequency

Probabilities as relative frequencies in the population: consider the Maryland Pick 3 lottery

- \$1 to play
- Choose three numbers between 0 and 9 (e.g. 000 $\rightarrow$ 999)
- Win \$500 if the three numbers are picked exactly



31

Consider the probability of certain events:

- Theoretical probabilities: how often would one *expect* to see a particular integer come up in one of the positions? ($1/10$)
- Probability as relative frequency: how often does a number *actually come up*?



32

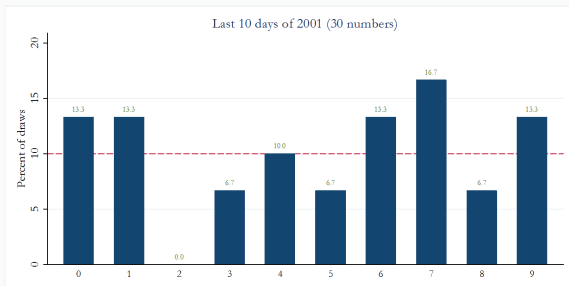
Probability as relative frequency

As the number of draws increases, the observed relative frequency should approach the true probabilities (assuming the game is fair).



33

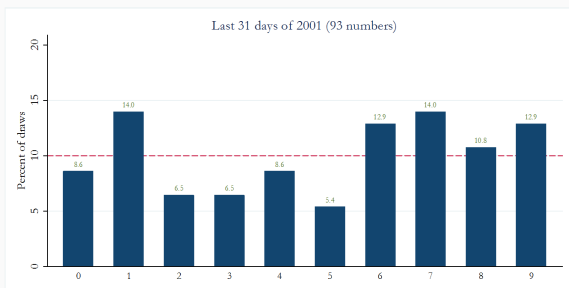
Probability as relative frequency



34



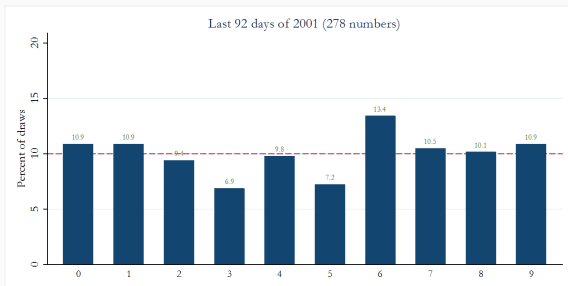
Probability as relative frequency



35



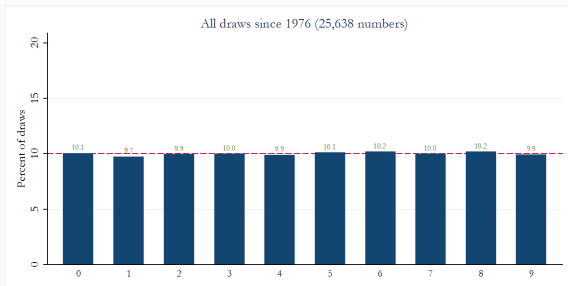
Probability as relative frequency



36



Probability as relative frequency



37



Probability distributions



Probability distributions

A **probability distribution** (or probability density) allows one to describe all of the possible outcomes of a random process and the probabilities those outcomes occur.

The form of the probability distribution function (or **PDF**) depends on whether the outcome is *discrete* or *continuous*.



Discrete probability distributions

For a **discrete** outcome, the probability distribution could just be a list of all possible outcomes and their probabilities. These probabilities must be positive and sum to 1.

- Analogous to the relative frequency distribution from Lecture 2.
- Simple example: # of heads in 3 coin flips (note there are 8 possible outcomes of 3 coin flips, when the order matters):

X	$P(X)$
0	1/8
1	3/8
2	3/8
3	1/8
SUM	1



39

Discrete probability distributions

The probability that X is between a and b , inclusive, is:

$$P(a \leq X \leq b) = \sum_{i=a}^b P(X = i)$$

For example, using the PDF above:

$$P(2 \leq X \leq 3) = P(X = 2) + P(X = 3) = (3/8) + (1/8) = 0.50$$



40

For **continuous** outcomes, a probability distribution function assigns probabilities to *intervals* of values.

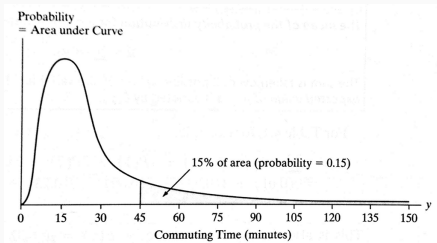
- Continuous random variables take on *infinitely many* outcomes
- The probability such a variable takes on a *specific* value is **zero**.
- The probability of the outcome falling in the interval containing *all* possible values (the sample space) is **one**.



41

Continuous probability distributions

When graphed, the **area** under a continuous probability distribution represents the probability that an outcome will fall within a particular interval:



42

Continuous probability distributions

Some will recognize that the area under a continuous function $f(x)$ between a and b can be found as:

$$\int_a^b f(x)dx$$

The area under $f(x)$ from $-\infty$ to $+\infty$ must be 1 to be a valid PDF.

Depending on the PDF, these areas can be obtained from formulas, tables, calculators, Stata, etc.



43

Discrete vs. continuous distributions



44

- How the function assigns a probability to every value of x
- “First principles”: what kind of a process lends itself to such a distribution? How does the distribution connect to the real world and the substantive problem at hand?
- How to use the final expression for the PDF.
- How to *simulate* random draws from the PDF.
- How to calculate features of the distribution such as its expected value/mean and variance (its **moments**).



Discrete probability distributions



Discrete probability distribution: Bernoulli

A **Bernoulli** distribution is the simplest discrete PDF, for outcomes that are binary 0-1 (like the gender of a newborn child). If $P(X) = P(\text{boy}) = 0.514$:

X	$P(X)$
0	0.486
1	0.514

Generally if $P(X = 1) = \pi$, then we can write:

$$P(X) = \pi^X(1 - \pi)^{(1-X)} \text{ for } X = 0, 1$$



46

Discrete probability distribution: binomial

Consider the possible gender combinations of 3 children had in sequence, where the order matters and births are independent events. $P(B)=0.514$ and $P(G)=0.486$.

Sequence	Probability
BBB	0.136
BBG	0.128
BGB	0.128
GBB	0.128
BGG	0.121
GGB	0.121
GBG	0.121
GGG	0.115
SUM	1.000



47

The above PDF can be written as follows if we are only interested in the probability of having k boys in 3 births:

# of boys	Probability
0	0.1148
1	0.3642
2	0.3852
3	0.1358
SUM	1.0000

This is an example of a **binomial distribution**.



48

Discrete probability distributions: binomial

The binomial distribution is useful with repeated *independent* and *identical* trials where the outcome of each trial is 0-1.

Define $X = 1$ generically as “success”. If $\pi = P(X = 1)$ on each trial, then the probability of k successes in n trials is:

$$P(k) = \binom{n}{k} \pi^k (1 - \pi)^{n-k}$$

where:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

is the number of ways to select k elements from a possible of n objects where the order does not matter. Note $n! = n * (n - 1) * (n - 2) * \dots * 1$. It is “ n choose k ” or ${}_nC_k$.



49

Examples: count data

- How many patients in a random sample of size n will experience side effects if the probability of side effects is π ?
- If π represents the proportion of emails that are spam, how many spam emails would you expect to receive in a day of n emails?
- How many days of school will be missed in a class of size n where the probability of being absent is π ?



50

Expected value: discrete random variable

The **mean** or **expected value** of a discrete random variable X with N unique outcomes is denoted $E(X)$ or μ_X and given by:

$$E(X) = \sum_{i=1}^N x_i P(x_i)$$

That is, it is the weighted average of outcomes where the weights are the probabilities each outcome occurs. It is a measure of *central tendency* for a probability distribution.

In the example above of 3 births:

$$E(X) = (0 \times 0.1148) + (1 \times 0.3642) + (2 \times 0.3852) + (3 \times 0.1358) = 1.542$$

The mean number of boys for parents with 3 births is 1.542.



The **variance** of a discrete random variable X with N unique outcomes is denoted $Var(X)$ or σ_X^2 and given by:

$$Var(X) = E(X - E(X))^2 = \sum_{i=1}^N (x_i - E(X))^2 P(x_i)$$

That is, it is the weighted average of squared deviations of each outcome x_i from $E(X)$, where the weights are the probabilities each outcome occurs. It is a measure of *dispersion*.

The standard deviation $SD(X)$ or (σ) is the square root of the variance.



52

Mean and variance: Bernoulli

Recall the Bernoulli PDF $P(X) = \pi^X(1 - \pi)^{(1-X)}$ where X is either 0 or 1.

$$E(X) = 0 * P(0) + 1 * P(1) = \pi$$

In our newborn gender case, $E(X) = 0.514$. The mean value of X is 0.514. (Or, 51.4% of newborns in the population are boys).

$$Var(X) = E(X - E(X))^2 = [(0 - \pi)^2 * (1 - \pi)] + [(1 - \pi)^2 * (\pi)] = \pi(1 - \pi)$$

$$SD(X) = \sqrt{\pi(1 - \pi)}$$



53

Recall the binomial PDF $P(k) = \binom{n}{k} \pi^k (1 - \pi)^{n-k}$.

$$E(X) = n\pi$$

On average, in n trials there are $n\pi$ successes. In our 3 successive births example, on average a mother will have $3 * 0.514 = 1.542$ boys. (Same answer as earlier).

$$\text{Var}(X) = n\pi(1 - \pi)$$

$$\text{SD}(X) = \sqrt{n\pi(1 - \pi)}$$

The variance in the number of boys (out of 3 successive births) will be $3 * 0.514 * (1 - 0.514) = \mathbf{0.749}$. The std deviation is $\sqrt{0.749} = \mathbf{0.865}$.



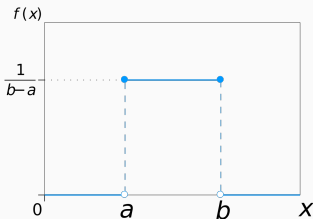
Continuous probability distributions



The uniform distribution

The **uniform (or rectangular) distribution** is a common continuous probability distribution function. Here:

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$



55



The uniform distribution

This works as a PDF since the area under this function between a and b is 1.

As a continuous PDF, the probability that X falls between values c and d (assuming $a \leq c \leq d \leq b$) is:

$$\int_c^d \frac{1}{b-a} dx$$

For the uniform distribution, this is just the area of a rectangle:

$$(d - c) * \frac{1}{b - a}$$

56



Think of this as the continuous equivalent of “equally likely outcomes” over a range $[a, b]$. It’s how you might model probability from a position of ignorance or lack of other information. Examples:

- Random number generator
- Guessing a birthday
- Arrival time of a bus



57

Expected value: continuous random variable

The **mean** or **expected value** of a continuous random variable X with PDF $f(x)$ is denoted $E(X)$ or μ_X and given by:

$$E(X) = \int_{-\infty}^{+\infty} xf(x)dx$$

The interpretation is analogous to the mean of a discrete random variable: it is a “weighting” of the possible x ’s, using the probability density as weights.



58

The **variance** of a continuous random variable X with PDF $f(x)$ is denoted $\text{Var}(X)$ or σ_X^2 and given by:

$$\text{Var}(X) = E(X - E(X))^2 = \int_{-\infty}^{+\infty} (x - E(X))^2 f(x) dx$$

The **standard deviation** (σ) is the square root of the variance.



59

Mean and variance: uniform

$$E(X) = \frac{1}{2}(a + b)$$

$$\text{Var}(X) = \frac{1}{12}(b - a)^2$$

$$SD(X) = \sqrt{\frac{1}{12}(b - a)^2}$$



60

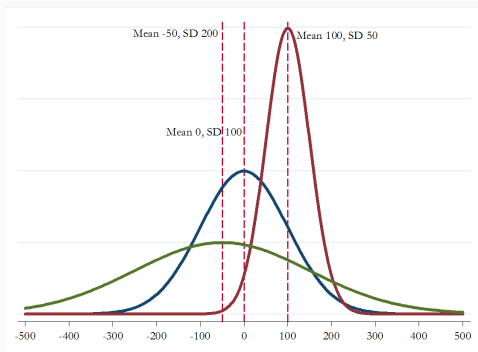
The normal distribution

- The **normal distribution** is probably the most important probability distribution in statistics (aka “Gaussian”)
- It is really a *family* of distributions, with different population means (μ) and standard deviations (σ)
- The normal distribution is *symmetric*:
 $P(x > \mu) = P(x < \mu) = 0.50$
- The area under the normal distribution is 1.



61

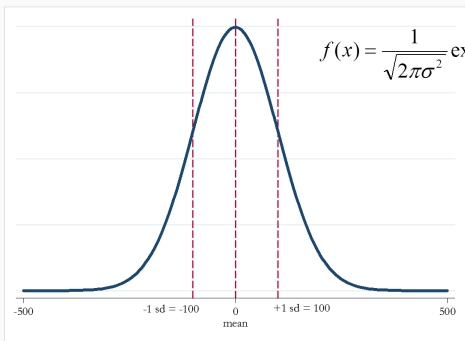
The normal distribution



62



The normal distribution



$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$$

63

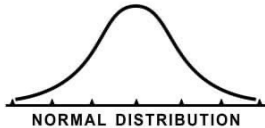


The normal distribution



64





65

The normal distribution

When a variable X has a normal distribution:

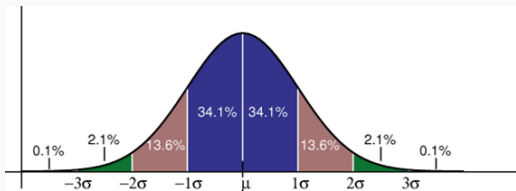
- The area between two values a and b is the probability that a randomly drawn X will fall within that interval: $P(a < X < b)$
- The area from $-\infty$ to k is the probability that a randomly drawn X will fall below k : $P(X < k)$
- The area from k to $+\infty$ is the probability that a randomly drawn X will fall above k : $P(X > k)$



66

The normal distribution

Recall the empirical rule: holds exactly for all normal distributions



67



The normal distribution

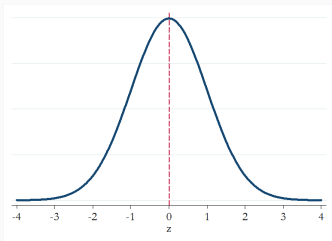
- Probabilities from the normal distribution can be obtained using Stata or an online calculator. Historically, they were taken from tables (as in the back of a textbook). You should know how to obtain probabilities using multiple methods.
- How would a table of probabilities work? As we saw, there are as many normal distributions as there are values of μ and σ .
- Solution: all normally distributed variables can be converted to **standard normal**—a z-score

68



The standard normal distribution

If X has a normal distribution, then its z -score: $z = (X - \mu)/\sigma$ has a **standard normal distribution** with a mean of zero and s.d. of 1:



69



The standard normal distribution

The standard normal makes computing probabilities easier:

$$P(X > x_1) = P\left(\frac{X - \mu}{\sigma} > \frac{x_1 - \mu}{\sigma}\right) = P\left(z > \frac{x_1 - \mu}{\sigma}\right)$$

70



The standard normal distribution: example

Suppose X has a normal distribution with a mean of 5 and a standard deviation of 2. What is the probability $X < 7.5$?

- It is common to write: $X \sim N(\mu, \sigma)$ or here $X \sim N(5, 2)$
- $P(X < 7.5) = P((X - 5)/2 < (7.5 - 5)/2) = P(z < 1.25)$
- Note $(7.5 - 5)/2 = 1.25$ is the z -score corresponding to $X = 7.5$ in the original distribution. 7.5 is 1.25 standard deviations above the mean.
- $P(z < 1.25) = 0.8944$ using an online calculator, Stata, or a lookup table.



71

The standard normal distribution: example

To compute probabilities from the standard normal distribution in Stata:

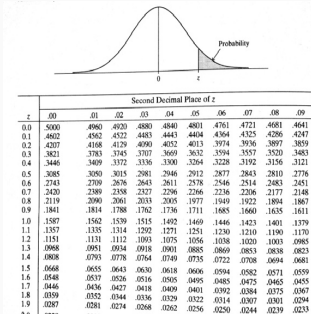
- The function `normal(z)` provides the probability of falling below z (the *cumulative probability*, or left tail)
- Example: `display normal(1.96)` yields 0.9750021
- Note `display 1-normal(z)` provides the probability of falling above z , the right tail



72

The standard normal distribution

A typical standard normal lookup table:

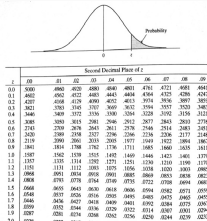


73

The standard normal distribution

Verifying the Empirical Rule using a standard normal distribution:
 $X \sim N(0, 1)$

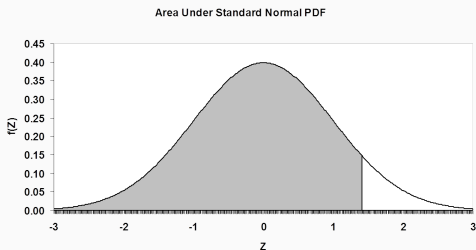
- $P(z > 2) = P(z < -2) = 0.021 + 0.001 = 0.022$
- $P(-1 < z < 1) = 0.341 + 0.341 = 0.682$
- $P(-2 < z < 2) = 0.136 + 0.341 + 0.341 + 0.136 = 0.954$



74

The standard normal distribution: examples

$$P(z < 1.41) = 0.9207$$

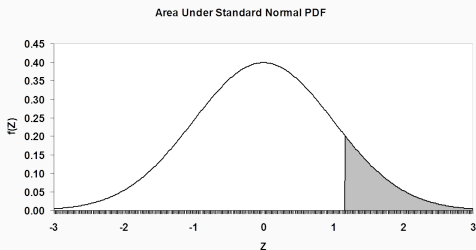


75



The standard normal distribution: examples

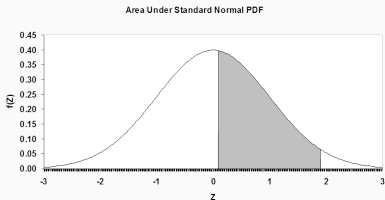
$$P(z > 1.17) = 1 - P(z < 1.17) = 1 - 0.8790 = 0.1210$$



76



$$P(0.1 < z < 1.9) = P(z < 1.9) - P(z < 0.1) = 0.9713 - 0.5398 = 0.4315$$



77



The standard normal distribution: examples

The time to complete an exam (X) is normally distributed with a mean of 110 and a standard deviation of 20. What is the fraction of students who will *not* finish in 120 minutes?

- $X \sim N(110, 20)$
- $P(X > 120) = 1 - P(X < 120)$
- $P(X < 120) = P((X - 110)/20 < (120 - 110)/20)$
- $1 - P(z < 0.5) = 1 - 0.6915 = 0.3085$

78



It is also possible to find the z value corresponding to a particular tail probability:

- The function `invnormal(p)` provides the z value for which the probability of falling *below* z is p
- Example: `display invnormal(0.025)` yields -1.96
- The probability of falling 1.96σ or more below the mean *or* 1.96σ or more above the mean is $2 \times 0.025 = 0.05$



79

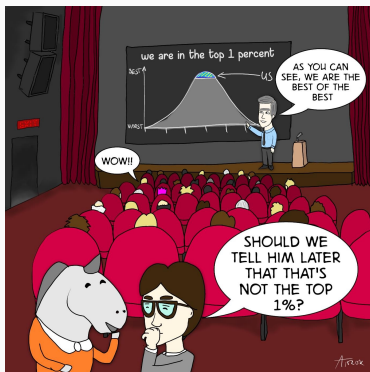
The standard normal distribution: examples

The time to complete an exam (X) is normally distributed with a mean of 110 and a standard deviation of 20. Above what value will X fall 25% of the time?

- $X \sim N(110, 20)$
- First find the value above which z will fall 25% of the time ($z = 0.6745$)
- The value of X that is 0.6745 standard deviations above its mean is: $110 + (0.6745 \times 20) = 123.49$



80



81



Simulating draws from probability distributions in Stata

Random number generators—like the one in Stata—are really *pseudo*-random number generators. That is, they are perfectly predictable mathematical functions.

If you know the pseudo-random number generator and *seed*, you will get the same sequence of numbers every time.

In Stata, `set seed #` where # is a number you choose.

This will allow you (and others) to replicate your random draws.

82



Simulating draws from probability distributions in Stata

Start from an empty dataset, and choose the number of observations you wish to generate:

```
clear
```

```
set seed 1989
```

```
set obs 1000
```

To generate 1000 draws from the $\text{uniform}(0,1)$ distribution:

```
gen x1=runiform()
```

To generate 1000 draws from the $\text{uniform}(a,b)$ distribution:

```
gen x2=runiform(a,b)
```

To generate 1000 draws from the $\text{binomial}(n,p)$ distribution—returns integers:

```
gen x3=rbinomial(n,p)
```



83

Simulating draws from probability distributions in Stata

To generate 1000 draws from the $\text{normal}(\mu, \sigma)$ distribution:

```
gen x4=rnormal(m, s)
```

To generate 1000 draws from the standard normal distribution:

```
gen x5=rnormal()
```

To generate 1000 draws from the Bernoulli distribution, use the $\text{uniform}(0,1)$ to generate random numbers and then assign the Bernoulli value (0 or 1) based on the result. For example, if $\pi = 0.514$:

```
gen x6=runiform()<0.514
```

Note this syntax is equivalent to saying “set x6 equal to 1 if `runiform()` is less than 0.514, and equal to 0 otherwise.”



84

See the handout on Github “Useful Stata commands for simulation”
(under Stata resources).



Next time



- Sampling variation and sampling distributions
- Standard errors
- Estimators and estimator properties
- More simulation: sampling distributions

